



7268 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
HR8799				
	1	NIRSpec-IFU-HR8799-Roll1	NIRSpec IFU Spectroscopy	(1) HR-8799
	7	NIRSpec-IFU-HR8799-Roll2	NIRSpec IFU Spectroscopy	(1) HR-8799
HD217697-Reference-Observation				
	6	NIRSpec-IFU-PSF-Ref	NIRSpec IFU Spectroscopy	(2) HD-218261-PSF-Reference

ABSTRACT

HR 8799 is the most iconic of the directly imaged exoplanetary systems. With four planets orbiting a single host star, it offers both a unique opportunity for comparative exoplanetology, and insight into the early history of a scaled-up version of our own solar system. Even with nearly two

decades of study, there remains tension in both measurements of the atmospheric properties of the planets and uncertainty into how this peculiar system formed. Ground-based observations produce incompatible spectroscopic measurements due to contamination from atmospheric turbulence and differences in instrumental calibration, making interpretation of the planetary spectra difficult. We therefore propose to obtain high SNR observations using NIRSpec/G140H and G235H, resolving the tension between the different spectroscopic observations, as well as between atmospheric measurements and predictions from formation models. While other JWST programs will trace the disequilibrium chemistry and cloud properties in the mid-infrared, this near-infrared window is critical to inferring basic properties such as the bolometric luminosity, metallicity, surface gravity, and carbon-to-oxygen number ratio. Without these parameters, interpretation of the existing JWST spectra will be biased, as they will rely on existing ground-based measurements, or model-dependent predictions of the near-infrared spectra. The HR 8799 planets are a benchmark system for direct imaging, but it is of fundamental importance that such a benchmark has reliable measurements, which can only be obtained through these JWST observations.

OBSERVING DESCRIPTION

HR 8799 is a benchmark system for directly imaged exoplanets, consisting of four giant planets orbiting a single host star. This system has seen extensive characterisation, however there remains significant tension between both ground-based spectra obtained with different instruments and between different measurements of atmospheric properties. We therefore propose to measure the near-infrared spectra of the four companions using NIRSpec G140H and G235H, which cover critical wavelength regions for measuring fundamental atmospheric properties of surface gravity, metallicity, carbon-to-oxygen number ratio, as well as the peak of the spectral energy distribution for the planets. This is a high-contrast observation, and we will therefore obtain a reference PSF observation of HD 21861 as well (the same reference as used in previous JWST observations of this system). Due to the brightness of the host star, saturation is unavoidable and our observing strategy reflects the need to both minimise saturation and obtain reasonable SNR for each of the companions.

Proposal 7268 - Targets - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HR-8799	RA: 23 07 28.8754 (346.8703142d) Dec: +21 08 2.61 (21.13406d) Equinox: J2000	Proper Motion RA: 108.284 mas/yr Proper Motion Dec: -50.03999999644293 mas/yr Parallax: 0.024462" Epoch of Position: 2016	
	<i>Comments: This object was generated based on Gaia DR3 coordinates from the Gaia archive, hence the 2016 epoch.</i> <i>It is not centered on the star, but the half-way point between HR 8799 b and HR 8799 d.</i> <i>Category=Star</i> <i>Description=[A stars, Exoplanet Systems, F stars]</i> <i>Extended=NO</i>				
	(2)	HD-218261-PSF-Reference	RA: 23 06 31.8853 (346.6328554d) Dec: +19 54 39.07 (19.91085d) Equinox: J2000	Proper Motion RA: 286.89 mas/yr Proper Motion Dec: 5.057 mas/yr Parallax: 0.03407719999999995" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[F stars]</i> <i>Extended=NO</i>				
	(3)	PSF-Ref-TA	RA: 23 06 31.5300 (346.6313750d) Dec: +19 54 37.95 (19.91054d) Equinox: J2000	Proper Motion RA: 281.902 mas/yr Proper Motion Dec: -1.4089999694988364 mas/yr Parallax: 0.0339839" Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Unidentified</i> <i>Description=[Visible sources]</i> <i>Extended=NO</i>				
	(5)	HR-8799-TA	RA: 23 07 27.9105 (346.8662938d) Dec: +21 08 41.82 (21.14495d) Equinox: J2000	Proper Motion RA: -4.277012105896459 mas/yr Proper Motion Dec: -5.578665846822867 mas/yr Parallax: 0.0005413746448693566" Epoch of Position: 2016.0	
<i>Comments:</i> <i>Category=Unidentified</i> <i>Description=[Visible sources]</i> <i>Extended=NO</i>					

Proposal 7268 - Observation 1 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Observation	Proposal 7268, Observation 1: NIRSpec-IFU-HR8799-Roll1										Thu Nov 13 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(NIRSpec-IFU-HR8799-Roll1 (Obs 1)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 69.575 Arcsec (larger than the recommended limit of 50.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
(NIRSpec-IFU-HR8799-Roll1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	HR-8799	RA: 23 07 28.8754 (346.8703142d) Dec: +21 08 2.61 (21.13406d) Equinox: J2000			Proper Motion RA: 108.284 mas/yr Proper Motion Dec: -50.03999999644293 mas/yr Parallax: 0.024462" Epoch of Position: 2016						
Comments: This object was generated based on Gaia DR3 coordinates from the Gaia archive, hence the 2016 epoch.												
It is not centered on the star, but the half-way point between HR 8799 b and HR 8799 d.												
Category=Star												
Description=[A stars, Exoplanet Systems, F stars]												
Extended=NO												
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	5 HR-8799-TA	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	219466.22	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	8	3	false	true	NONE	16	48	4638.285	

Proposal 7268 - Observation 1 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Special Requirements	Aperture PA Range 193.97164917 to 213.97164917 Degrees (V3 55.0 to 75.0) Group Observations 1, 7, Non-interruptible Group Observations 1, 6, 7 within 1 Days Aperture PA Offset 7 from 1 by 10 to 16 Degrees (Same offsets in V3) Same Aperture PA 1, 6
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Proposal 7268 - Observation 7 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Observation	Proposal 7268, Observation 7: NIRSpec-IFU-HR8799-Roll2										Thu Nov 13 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(NIRSpec-IFU-HR8799-Roll2 (Obs 7)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 69.575 Arcsec (larger than the recommended limit of 50.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(NIRSpec-IFU-HR8799-Roll2 (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	HR-8799	RA: 23 07 28.8754 (346.8703142d) Dec: +21 08 2.61 (21.13406d) Equinox: J2000			Proper Motion RA: 108.284 mas/yr Proper Motion Dec: -50.03999999644293 mas/yr Parallax: 0.024462" Epoch of Position: 2016						
	Comments: This object was generated based on Gaia DR3 coordinates from the Gaia archive, hence the 2016 epoch.											
	It is not centered on the star, but the half-way point between HR 8799 b and HR 8799 d.											
	Category=Star Description=[A stars, Exoplanet Systems, F stars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	5 HR-8799-TA	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	219466.22	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	8	3	false	true	NONE	16	48	4638.285	

Proposal 7268 - Observation 7 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Special Requirements	Aperture PA Range 193.97164917 to 213.97164917 Degrees (V3 55.0 to 75.0) Group Observations 1, 7, Non-interruptible Group Observations 1, 6, 7 within 1 Days Aperture PA Offset 7 from 1 by 10 to 16 Degrees (Same offsets in V3)
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Proposal 7268 - Observation 6 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Observation	Proposal 7268, Observation 6: NIRSpec-IFU-PSF-Ref										Thu Nov 13 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(NIRSpec-IFU-PSF-Ref (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	HD-218261-PSF-Reference	RA: 23 06 31.8853 (346.6328554d) Dec: +19 54 39.07 (19.91085d) Equinox: J2000			Proper Motion RA: 286.89 mas/yr Proper Motion Dec: 5.057 mas/yr Parallax: 0.03407719999999995" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[F stars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	3 PSF-Ref-TA	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	219466.22	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	8	3	false	true	NONE	16	48	4638.285	

Proposal 7268 - Observation 6 - Easing the Tension: Near-Infrared Spectroscopy of HR 8799 b,c,d,e

Special Requirements	Group Observations 1, 6, 7 within 1 Days Same Aperture PA 1, 6
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